

ECONOMETRICS I
Problem Set 2: OLS applications and variance

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November 23, 2025

1. **Application: real estate data.** Download `br.RData` from StudIP. This dataset (originally from Hill et al. 2018, *Principles of Econometrics*) contains data on 1080 houses sold in Baton Rouge, Louisiana, during mid-2005.
 - (a) Plot `price` vs. `sqft`. What kind of model or transformation would the scatter suggest?
 - (b) Estimate the relationship of `price` vs. `sqft` with a level-level, a quadratic polynomial, a log-level, and a log-log model. You should get the results shown in table 1. What model offers the best fit? Does your graphical analysis lead to the same conclusion?
 - (c) You are considering buying a 2,000 square-foot property for \$ 120,000. Is that a good price? Answer according to each estimate of table 1.
 - (d) Interpret the effect of size on price according to each estimate of of table 1.
 - (e) Do the regressions exhibit homoskedastic errors? Generate the residuals via, e.g., `br$resid1 <- residuals(model1)` and plot them against `sqft`. What is no longer valid in case of heteroskedasticity?

2. **R-squared**

- (a) Explain graphically why the variance of y can be decomposed as the variance of $\hat{y}$ and that of $\hat{u}$.
- (b) Show that

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{TSS} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{ESS} + \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{SSR}$$

so that $0 \leq R^2 \leq 1$.

3. **Unbiasedness.** We say that an estimator $\hat{\theta}$ of the population parameter θ is unbiased if, on average in repeating sampling, it coincides with its population counterpart—i.e.,

Table 1: Result of question 1b

	<i>Dependent variable:</i>			
	price		log(price)	
	(1)	(2)	(3)	(4)
sqft	92.747*** (2.411)	-41.153*** (6.623)	0.0004*** (0.00001)	
I(sqft^2)		0.021*** (0.001)		
log(sqft)				1.006*** (0.025)
Constant	-60,861.460*** (6,110.187)	113,226.500*** (9,672.195)	10.839*** (0.025)	4.080*** (0.189)
Observations	1,080	1,080	1,080	1,080
R ²	0.579	0.703	0.625	0.607

Note:

*p<0.1; **p<0.05; ***p<0.01

when $\mathbb{E}(\hat{\theta}) = \theta$.

- (a) Distinguish graphically between unbiasedness and efficiency.
 (b) Starting from the formula

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

of the simple (univariate) regression model, show that the condition $\mathbb{E}(u_i|x) = 0$ —or, alternatively, $\mathbb{E}[\text{cov}(u, x)] = 0$ —is required in addition to the assumption

$$y = \beta_1 + \beta_2 x + u$$

for $\mathbb{E}(\hat{\beta}_2|x) = \beta_2$ to hold.

- (c) Starting from the formula

$$\hat{\beta} = (X'X)^{-1}X'y$$

of the multivariate regression model, show that the condition $\mathbb{E}(u|X) = 0$ —or, alternatively, $\mathbb{E}(X'u) = 0$ —is required in addition to the assumption

$$y = X\beta + u$$

for $\mathbb{E}(\hat{\beta}|X) = \beta$ to hold.

4. **Estimator variance.** We define an estimator variance as $\mathbb{E}[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2]$ in repeating sample.

- (a) Show that the variance of the slope $\hat{\beta}_2$ in a simple regression that we obtained in class,

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (1)$$

can also be obtained from solving $\mathbb{E}(\hat{\beta}_2 - \mathbb{E}(\hat{\beta}_2))^2$.

- (b) Show that, for a random vector $a_{(n \times 1)}$, it holds that

$$V[a] \equiv \mathbb{E}[(a - \mathbb{E}[a])(a - \mathbb{E}[a])'] = \begin{bmatrix} \text{var}[a_1] & \text{cov}[a_1, a_2] & \cdots & \text{cov}[a_1, a_n] \\ \text{cov}[a_2, a_1] & \text{var}[a_2] & \cdots & \text{cov}[a_2, a_n] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[a_n, a_1] & \text{cov}[a_n, a_2] & \cdots & \text{var}[a_n] \end{bmatrix}$$

$n \times n$

Note: we call $V[a]$ the **variance-covariance matrix** of a .

- (c) Show that the assumptions of homoskedasticity and no-autocorrelation of the error can be summarized as

$$V[u|X] = \mathbb{E}[uu'|X] = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_{n \times n} \quad (2)$$

- (d) Show that the matrix analog of (1) is

$$V(\hat{\beta}|X) = \begin{bmatrix} \text{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \text{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2 (X'X)^{-1}$$

$k \times k$

- (e) Make an R script computing $\hat{V}[\hat{\beta}|X] = \hat{\sigma}^2 (X'X)^{-1}$, where $\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k} = \frac{\hat{u}'\hat{u}}{n-k}$.